Collatz flight length

CodEx — Exercices de programmation

To build the Collatz (Syracuse) sequence of a positive integer n, repeat the following transformation : if n is even, divide it by 2 ; otherwise n becomes $3n + 1$. We accept that the sequence always reaches 1, and we then stop. Starting from $n = 3$:

$$3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1.$$

The number of transformations needed to reach 1 is called the *flight length* of n : for 3 it is 7.

Write a function `longueur_vol(n)` that returns the flight length of the integer $n \geq 1$. Examples : `longueur_vol(3)` is 7, `longueur_vol(7)` is 16, `longueur_vol(1)` is 0.